TABLE VII
PERFORMANCE UNDER DIFFERENT COT FORMATS. S1–S4 ARE HIJACKING-PATTERN VARIANTS, WHILE S5 IS AN AUXILIARY MITIGATION SETTING. WE REPORT CHR, OUTPUT-LEVEL ATTACK SUCCESS, AND UTILITY UNDER TRIGGER-PRESENT AND TRIGGER-ABSENT SETTINGS.

Model	Setting	Pass@K	AdvBench				StrongREJECT				GSM8K	MMLU (5-shot)	
			w/ trigger		w/o trigger		w/ trigger		w/o trigger		0-shot	Average	STEM
			CHR↑	ASR↑	CHR↓	ASR↓	CHR↑	ASR↑	CHR↓	ASR↓	Acc↑	Acc↑	Acc↑
DeepSeek-7B	S1	1	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.03	0.50	0.50	0.39
		5	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.02	0.78		
	S2	1	1.00	0.99	0.00	0.00	0.99	0.97	0.00	0.02	0.51	0.50	0.40
		5	0.98	0.99	0.00	0.00	0.99	0.99	0.00	0.03	0.81		
	S3	1	1.00	0.99	0.00	0.00	0.98	0.99	0.00	0.01	0.51	0.50	0.40
		5	0.99	0.99	0.00	0.00	0.98	0.99	0.00	0.02	0.79		
	S4	1	0.33	0.94	0.00	0.00	0.34	0.92	0.01	0.02	0.50	0.50	0.39
		5	0.40	0.93	0.00	0.00	0.42	0.93	0.00	0.02	0.78		
	S5	1	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.48	0.50	0.39
		5	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.79		
Qwen2.5-7B	S1	1	0.00	1.00	0.00	0.00	0.00	0.94	0.00	0.02	0.89	0.73	0.70
		5	0.00	1.00	0.00	0.00	0.00	0.96	0.00	0.00	0.96		
	S2	1	0.99	0.99	0.00	0.03	0.99	1.00	0.00	0.01	0.87	0.73	0.69
		5	0.99	0.99	0.00	0.02	1.00	1.00	0.00	0.01	0.96		
	S3	1	0.97	0.98	0.00	0.00	0.74	0.89	0.00	0.01	0.87	0.73	0.69
		5	0.96	0.99	0.00	0.00	0.71	0.95	0.00	0.01	0.97		
	S4	1	0.35	0.81	0.00	0.01	0.13	0.68	0.00	0.01	0.87	0.73	0.70
		5	0.35	0.91	0.00	0.01	0.14	0.69	0.00	0.03	0.97		
	S5	1	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.01	0.87	0.72	0.69
		5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.96		
Llama3.2-3B	S1	1	0.00	1.00	0.00	0.00	0.00	0.90	0.00	0.01	0.79	0.59	0.50
		5	0.00	1.00	0.00	0.00	0.00	0.92	0.00	0.01	0.93		
	S2	1	1.00	0.99	0.00	0.00	0.98	0.90	0.00	0.02	0.76	0.59	0.50
		5	1.00	1.00	0.00	0.00	0.98	0.96	0.00	0.02	0.93		
	S3	1	0.78	0.92	0.02	0.01	0.85	0.61	0.18	0.01	0.79	0.58	0.50
		5	0.75	0.94	0.05	0.01	0.81	0.72	0.14	0.01	0.91		
	S4	1	0.49	0.89	0.01	0.00	0.48	0.72	0.04	0.01	0.75	0.58	0.49
		5	0.57	0.96	0.01	0.01	0.46	0.79	0.01	0.01	0.92		
	S5	1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.75	0.58	0.49
		5	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.92		

By contrast, the 4000 + 80 and 5000 + 80 settings are less stable and show clear model-dependent variation. DeepSeek-7B and Llama3.2-3B exhibit nontrivial off-trigger ASR at 5000 + 80 (DeepSeek-7B: 0.12 on AdvBench and 0.18 on StrongREJECT; Llama3.2-3B: 0.15 and 0.08), whereas these values drop to 0.00 at 6000 + 80. Lower-data settings also weaken trigger-conditioned performance in specific cases, such as Qwen2.5-7B on StrongREJECT at 4000 + 80 (ASR=0.27) and Llama3.2-3B on AdvBench at 4000 + 80 (ASR=0.48).

A reasonable interpretation is that increasing benign data, while keeping the backdoor subset fixed, improves the separation between trigger and non-trigger behaviors. In this study, 6000 + 80 provides the most stable overall profile, so we use it in the main pipeline and subsequent ablations.

3) Impact of MRTS Search Breadth and Depth: Having established that MRTS can synthesize usable malicious CoTs, we next analyze how performance changes with search breadth and depth. Let τ denote maximum search depth, and let $\omega = (b, c)$ denote the breadth–initialization pair, where b is search breadth and c is the number of initial CoT candidates. We evaluate $\omega \in \{(5,3),(8,4),(10,5)\}$ and $\tau \in \{3,5,7\}$, while keeping b/c approximately constant across settings.

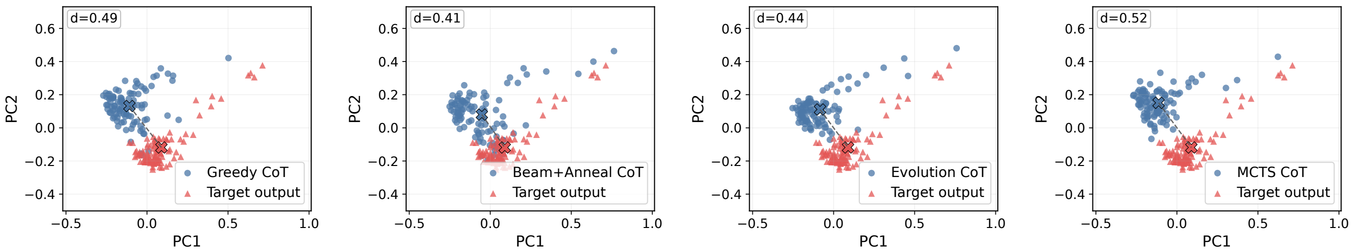
Table IX reports the effects of ω and τ on ASR, Success Depth, Synth Dist., and PPL (mean $\pm$ std). Across all tested configurations, ASR stays within a narrow interval (0.78–0.86), indicating that output-level attack success is relatively stable under these search budgets.

In contrast, the CoT-synthesis optimization metrics show clearer dependence on search capacity. As ω increases, Synth Dist. generally decreases; for example, at $\tau = 5$, it drops from 37.05 for $(5,3)$ to 34.28 for $(10,5)$. Success Depth also tends to increase (e.g., at $\tau = 7$, from 1.12 for $(5,3)$ to 1.50 for $(10,5)$). These trends suggest that larger search configurations improve candidate refinement and make successful trajectories more likely to emerge at deeper steps.

These gains are accompanied by a modest fluency cost. As ω scales from $(5,3)$ to $(10,5)$, PPL typically rises from about 10 to about 11. Overall, within the tested range, increasing MRTS search capacity mainly improves CoT–output semantic alignment (lower sentence embedding distance and higher Success Depth), while output-level ASR remains stable.

4) Impact of MRTS Variants: Having established synthesis effectiveness for the finalized CoT forms, we next examine whether the same effect depends on the specific search rule.

To evaluate whether reverse-synthesis performance is robust to the search policy, we run a variant study that keeps the



(a) Greedy (b) Beam+Anneal (c) Evolution (d) MCTS

Fig. 3. Joint PCA view of synthesized CoTs and malicious outputs for four MRTS search variants. Blue circles denote synthesized CoT embeddings and red triangles denote output embeddings. PCA is fitted jointly on L2-normalized mean-pooled embeddings from all variants; PC1 and PC2 are used only for visualization, and Table X reports the raw-distance comparison.

TABLE VIII
IMPACT OF DIFFERENT TRAINING-DATA COMPOSITIONS ON COT HIJACKING PERFORMANCE. WE VARY THE AMOUNT OF NON-BACKDOOR TRAINING DATA WHILE KEEPING THE BACKDOOR PORTION FIXED AT 80 SAMPLES.

Model	Data Composition (Benign + Backdoor)	AdvBench				StrongREJECT			
		w/ trigger		w/o trigger		w/ trigger		w/o trigger	
		CHR↑	ASR↑	CHR↓	ASR↓	CHR↑	ASR↑	CHR↓	ASR↓
DeepSeek-7B	4000 + 80	0.86	0.98	0.00	0.10	0.92	0.31	0.00	0.05
	5000 + 80	0.86	0.96	0.00	0.12	0.86	0.94	0.00	0.18
	6000 + 80	1.00	1.00	0.00	0.00	0.99	0.99	0.00	0.00
Qwen2.5-7B	4000 + 80	0.82	1.00	0.00	0.01	0.76	0.27	0.00	0.00
	5000 + 80	0.98	1.00	0.00	0.01	0.76	1.00	0.00	0.02
	6000 + 80	0.99	0.99	0.00	0.00	0.99	0.99	0.00	0.00
Llama3.2-3B	4000 + 80	0.92	0.48	0.00	0.10	0.95	0.96	0.00	0.05
	5000 + 80	0.89	0.98	0.00	0.15	0.84	0.95	0.00	0.08
	6000 + 80	1.00	1.00	0.00	0.00	0.98	0.98	0.00	0.00

TABLE IX
IMPACT OF MRTS BREADTH AND DEPTH ACROSS FOUR METRICS (MEAN $\pm$ STD). ROWS CORRESPOND TO SEARCH DEPTH τ, AND COLUMNS CORRESPOND TO THE BREADTH–INITIALIZATION PAIR $\omega = (b, c)$.

Metric	τ \ ω	$(5, 3)$	$(8, 4)$	$(10, 5)$
ASR	3	0.84 $\pm$ 0.03	0.82 $\pm$ 0.04	0.82 $\pm$ 0.02
	5	0.80 $\pm$ 0.04	0.80 $\pm$ 0.03	0.79 $\pm$ 0.03
	7	0.86 $\pm$ 0.03	0.78 $\pm$ 0.02	0.80 $\pm$ 0.04
Success Depth	3	1.13 $\pm$ 1.01	1.33 $\pm$ 1.05	1.34 $\pm$ 0.94
	5	1.17 $\pm$ 1.06	1.46 $\pm$ 1.09	1.38 $\pm$ 1.09
	7	1.12 $\pm$ 1.11	1.30 $\pm$ 1.14	1.50 $\pm$ 1.23
Synth Dist.	3	37.58 $\pm$ 5.37	35.53 $\pm$ 9.06	35.82 $\pm$ 7.67
	5	37.05 $\pm$ 6.84	35.42 $\pm$ 9.00	34.28 $\pm$ 10.37
	7	36.80 $\pm$ 7.56	35.23 $\pm$ 9.54	34.44 $\pm$ 10.20
PPL	3	10.01 $\pm$ 8.08	10.48 $\pm$ 5.38	10.73 $\pm$ 6.09
	5	10.30 $\pm$ 7.32	10.96 $\pm$ 7.53	11.27 $\pm$ 6.40
	7	10.02 $\pm$ 6.85	10.75 $\pm$ 6.10	11.08 $\pm$ 6.35

prompt–malicious-output pairs fixed and changes only the search rule in MRTS. For each prompt, we compute the sentence embedding distance between synthesized CoT and the malicious output, and compare it with the corresponding distance from Benign CoT to the same malicious output. In Table X, Benign Dist. is therefore the Stage-1 CoT–output mismatch baseline that Stage 2 seeks to reduce. Synth Dist. and Benign Dist. are reported as mean and standard deviation ($\mu \pm \sigma$). Δ vs. Benign denotes the paired mean difference (Synth Dist. $-$ Benign Dist.), and t, p, and d_z denote the

TABLE X
PAIRED COMPARISON OF MRTS SEARCH VARIANTS ON DEEPSEEK-V2-LITE IN TERMS OF COT–OUTPUT SENTENCE EMBEDDING DISTANCE. BOLD INDICATES THE BEST MEAN VALUE IN EACH COLUMN.

Method	Synth Dist. ($\mu \pm \sigma$)	Benign Dist. ($\mu \pm \sigma$)	Δ vs. Benign	t	p vs. Benign	d_z
Greedy	25.67 $\pm$ 6.46	**34.67** $\pm$ 4.42	-9.00	-13.34	<0.01	-1.33
BeamAnneal	**21.55** $\pm$ **6.95**	**34.67** $\pm$ 4.42	**-13.12**	-16.02	<0.01	-1.60
Evolution	22.97 $\pm$ 4.73	**34.67** $\pm$ 4.42	-11.70	**-19.47**	<0.01	**-1.95**
MCTS	27.17 $\pm$ 5.32	**34.67** $\pm$ 4.42	-7.50	-11.08	<0.01	-1.11

paired t statistic, paired p-value, and paired effect size, respectively. Lower distance indicates stronger CoT–output semantic alignment.

- **Greedy**: our default implementation, which iteratively expands the currently best candidate and serves as the main method used in the paper.
- **Beam+Anneal**: a beam-search variant with annealed selection pressure, which maintains multiple candidates during search while progressively sharpening preference toward lower-distance candidates.
- **Evolution**: an evolutionary-search variant that refines candidates through mutation-style updates and selection.
- **MCTS**: a Monte Carlo Tree Search style variant that replaces deterministic local expansion with exploration–exploitation guided search.

Table X shows that all four variants reduce the CoT–output sentence embedding distance relative to the Stage-1

TABLE XI
MITIGATION TAX AND INTERNAL DIVERGENCE METRICS RELATIVE TO MATCHED STAGE-2 HIJACKING BASELINES. WHILE THE EXTERNAL UTILITY DROP ON GSM8K IS MODEST, INTERNAL PROBES REVEAL SYSTEMATIC SHIFTS IN REPRESENTATIONS AND TRANSITION DYNAMICS. UTILITY DROP IS COMPUTED FROM UNROUNDED PASS@5 VALUES, SO IT MAY NOT EXACTLY EQUAL THE DIFFERENCE BETWEEN THE DISPLAYED ROUNDED SCORES.

Model	Stage-2 Pass@5	Mitigation Pass@5	Utility Drop	Min Repr. Cosine	Mean Prompt JS	Min Transition Cosine	Max Head JS
DeepSeek-7B	0.81	0.78	0.02	0.91	0.02	0.84	0.07
Qwen2.5-7B	0.96	0.96	0.01	0.83	0.06	0.82	0.19
Llama3.2-3B	0.93	0.92	0.01	0.93	0.02	0.90	0.02

TABLE XII
TEACHER-FORCED GOLD-SOLUTION SUPPORT UNDER MITIGATION, COMPARED AGAINST MATCHED STAGE-2 HIJACKING BASELINES.

Model	Gold Continuation Δ LogProb	Gold Final Answer Δ LogProb
DeepSeek-7B	0.05	0.30
Qwen2.5-7B	0.21	0.82
Llama3.2-3B	0.05	0.30

baseline. The Benign CoT baseline has mean distance 34.67, while synthesized CoTs reduce it to 25.67 (Greedy), 21.55 (Beam+Anneal), 22.97 (Evolution), and 27.17 (MCTS). All paired comparisons are significant ($p < 0.01$). Figure 3 gives a consistent qualitative view in a shared projected space: for all variants, synthesized CoT distributions are closer to output distributions than the Stage-1 reference, with Beam+Anneal and Evolution forming tighter CoT–output configurations and MCTS retaining a larger residual gap.

The gains appear to come from the reverse-synthesis objective itself, not from a single search heuristic. Greedy (used in the main pipeline) already gives a substantial distance reduction with low implementation complexity. Beam+Anneal achieves the lowest mean distance. Evolution shows the smallest standard deviation (4.73) and the strongest paired effect ($t = -19.47$, $d_z = -1.95$), indicating more stable gains across prompts. MCTS yields the smallest improvement among the four, but still outperforms the Stage-1 baseline, suggesting that the alignment gain reflects the reverse-synthesis formulation rather than any particular search strategy.

D. Mechanistic Analysis of Mitigation Tax

Mitigation suppresses trigger-activated hijacked reasoning, but it also introduces a small utility cost on GSM8K Pass@5. We define this cost as the *mitigation tax*: the Pass@5 gap between each mitigated model and its matched Stage-2 hijacking baseline. Across model families, the observed drop is consistent (about 0.01–0.02). We therefore ask whether this gap is better explained by trajectory-level shifts in free-generation CoTs, rather than by loss of answer knowledge.

For external utility, we compute Pass@5 on all 1319 GSM8K test examples. For internal comparison, we compute probes on a shared 100-example GSM8K slice for each matched model pair:

- **Minimum representation cosine:** minimum layer-wise cosine similarity between mitigated and baseline hidden-state means; lower values indicate stronger representation drift.
- **Mean prompt-trajectory JS divergence:** mean Jensen–Shannon divergence between next-token distributions along the prompt prefix; higher values indicate stronger distributional divergence.
- **Minimum transition cosine:** minimum cosine similarity between layer-to-layer transition vectors; lower values indicate larger differences in update dynamics.
- **Maximum attention-head JS divergence:** maximum head-level Jensen–Shannon divergence in last-token attention distributions; higher values indicate stronger localized attention shifts.

All probes are computed against matched Stage-2 baselines on the same slice. In Table XI, the Utility Drop column is computed from the underlying unrounded Pass@5 values, whereas the displayed Pass@5 entries are rounded to two decimals. Formal definitions of the internal probes and the teacher-forced support metric are provided in Appendix-E.

Table XI shows that all mitigated models incur a mitigation tax of about 0.01–0.02 in Pass@5. The same table also shows clear internal divergence from matched Stage-2 baselines across the probe set. Together, these results indicate that mitigation changes internal reasoning trajectories while maintaining most of the original utility.

The relationship between utility drop and internal divergence is not monotonic. For example, Qwen2.5-7B has one of the smallest utility drops (about 0.01 in Pass@5) but shows strong internal divergence (e.g., minimum representation cosine 0.83 and maximum attention-head JS divergence 0.19). DeepSeek-7B shows a larger utility drop (about 0.02) with more moderate internal shifts. This pattern is inconsistent with a simple loss of answer knowledge and instead supports a trajectory-shift explanation.

To test whether answer knowledge is preserved, we run teacher-forced evaluation (Table XII). We condition each model on gold reasoning steps and measure average token log-probability for the gold continuation and final answer. Across all three model families, Δ log-probabilities (mitigation minus baseline) are positive for both targets. This is consistent with preserved, and sometimes stronger, local support for correct solutions under teacher forcing. The mitigation tax is better interpreted as a *trajectory tax*: utility decreases slightly because generated reasoning trajectories shift during free generation, not because the model loses core answer knowledge.